

10.3 Groq API

If you would like to experiment with publicly available LLMs but don't have a sufficiently fast computer, Groq (<https://wow.groq.com>) is a very helpful alternative. This company currently offers developers a free option to run a few selected language models (llama, mixtral, gemma, whisper).

What is remarkable about Groq is the incredible speed at which prompts are answered—much faster than with good proprietary hardware but also significantly faster compared to the OpenAI API. This speed is especially great for experimenting!

Developer accounts are subject to rate limits of several thousand tokens per minute (for example, 20,000 tokens/minute for llama-3.1-8b-instant). Due to the high speed, however, these limits are quickly exceeded if you process multiple prompts in one script. Of course, we can't say how long this developer access will remain free as commercial use is always subject to a charge.

It's best to set up a Python environment directory for initial experiments. The following commands apply to Linux and macOS:

```
python3 -m venv groq
cd groq
source bin/activate
pip3 install groq
pip3 install python-dotenv
```

To use Groq, you need an API key. You'll receive this key after an uncomplicated login process (you only need to enter your email address) on the Groq website. You save the key in the `.env` file in the project directory:

```
# File .env
GROQ_API_KEY=gsk_XXX
```

The use of the API is very similar to that of OpenAI or Ollama:

```
# Sample file groq/hello.py
import os
from groq import Groq
from dotenv import load_dotenv

load_dotenv()
client = Groq(api_key=os.getenv("GROQ_API_KEY"))
completion = client.chat.completions.create(
    model="llama3-8b-8192",
    messages=[
        { "role": "user",
          "content": "How do I use list comprehension in
Python?" } ])

print("finish_reason:",
      completion.choices[0].finish_reason)
print("input token:",
      completion.usage.prompt_tokens)
print("output token:", completion.usage.total_tokens)
print()
print(completion.choices[0].message.content)
```

